CHOMP!

Wally the Cairns Crocodile

Game Design Document v1.0

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Powered By	**ButterflyFX**
Creative Director	**Kenneth Bingham**
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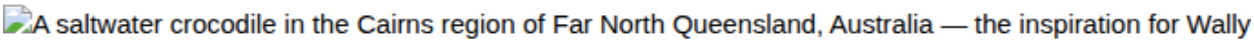
Genre	**2.5D Platformer / Action-Adventure**
Target Platforms	**Steam, Google Play, Major Game Platforms**

"Store Geometry Not Bytes"



Saltwater crocodile in the Cairns region — the real-world inspiration for Wally, our protagonist.

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1. Game Overview

High Concept

CHOMP! is a vibrant 2.5D platformer set in and around Cairns, Far North Queensland, Australia. Players control **Wally**, a charismatic saltwater crocodile navigating the tropical wilds, urban edges, and underwater reefs of the Cairns region. The game celebrates authentic regional culture, wildlife, humor, and hazards — all built on **ButterflyFX** manifold geometry.

Unique Selling Points

- **Authentic Cairns, Australia setting** with real regional wildlife, landmarks, and humor
- **Manifold-based procedural level architecture** powered by the ButterflyFX engine
- **Geometry-first asset pipeline** — "Store Geometry Not Bytes"
- **Rich cast of regionally authentic ANPCs** (Animated Non-Player Characters)
- **Multiple biomes** spanning rainforest, reef, mangrove, urban, and outback

Target Audience

Ages 10+, platformer fans, Australian culture enthusiasts, and the indie game community. CHOMP! is designed to appeal to both casual players seeking a charming, exploration-driven experience and core platformer fans looking for tight mechanics and satisfying mastery curves.

Game Pillars

Pillar	Description
Exploration	Rich, layered biomes reward curiosity with hidden paths, collectibles, and

Pillar	Description
	lore. Every corner of Cairns holds something to discover.
Humor	Dry Australian wit permeates dialogue, character interactions, and environmental storytelling. The game never takes itself too seriously.
Regional Authenticity	Real Cairns landmarks, wildlife, hazards, and culture are woven into every level. Players should feel like they've visited Far North Queensland.
Fluid Movement	Wally's movement set — from galloping runs to sinusoidal swimming — should feel powerful, responsive, and deeply satisfying.
Discovery	Environmental storytelling, ANPC side quests, and lore postcards layer narrative meaning over gameplay. The world tells its own story.

Design Philosophy

CHOMP! is built on the principle of **"Store Geometry Not Bytes."** ButterflyFX manifold surfaces encode level geometry as mathematical parameters rather than static mesh data. This enables dramatic file-size reductions, seamless procedural generation, and the organic, flowing terrain curves that define the Australian landscape.

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2. Story & Narrative

Premise

Wally is a young saltwater crocodile living in the estuaries near Cairns. When a mysterious environmental disruption begins affecting the local ecosystem — cane toads multiplying out of control, coral bleaching spreading, and sugar cane fields

being overtaken by invasive species — Wally sets out on an adventure across the region to restore balance.

Story Arcs

Act	Title	Setting	Description
Act 1	The Estuaries	Cairns Estuarine Wetlands	Wally discovers the disruption in his home territory. Tutorial levels introduce core mechanics — running, jumping, swimming, and the signature CHOMP! attack. Wally meets Baz the Barramundi, who sets him on his path.
Act 2	Into the Wild	Daintree Rainforest, Cane Fields, Mangroves	Wally travels through the Daintree Rainforest, mangrove swamps, and sugar cane fields. He encounters allies — Bruce the Cassowary, Shazza the Flying Fox — and regional hazards including cane toad swarms, feral pig stampedes, and a rival saltwater croc. Culminates in the Giant Cane Toad King boss fight.
Act 3	The Reef	Great Barrier Reef	Underwater levels on the Great Barrier Reef. Wally must navigate coral formations, jellyfish swarms,

Act	Title	Setting	Description
			and reef predators. Coral the Sea Turtle serves as guide. Breath management and aquatic combat take center stage. Boss: Box Jellyfish Colossus.
Act 4	Urban Edge	Cairns City Outskirts	Wally ventures into the outskirts of Cairns city. He encounters human infrastructure, storm drains, and the iconic Night Markets area. Stealth mechanics introduce a new dimension as Wally avoids detection. Maggie the Magpie provides streetwise intel.
Act 5	The Source	Atherton Tablelands	Wally traces the disruption to its origin deep in the Atherton Tablelands. Volcanic crater lakes, cascading waterfalls, and ancient terrain lead to the final boss arena. Old Terry the Taipan provides cryptic guidance. Final confrontation with The Disruptor.

Narrative Tone

Light-hearted Australian humor with environmental awareness undertones. The game employs regional slang, dry wit, and self-aware comedy throughout. Narrative

delivery is **humor-first** — there is no heavy-handed messaging. Environmental themes emerge naturally through gameplay and world-building rather than lecturing the player.

Sidequests

Sidequest	Description	Biome	Type
Toad Roundup	Collect and relocate cane toads from sensitive areas before they overrun native habitats.	Estuaries / Cane Fields	Collection
Flying Fox Rescue	Help displaced spectacled flying foxes find new roosts across the rainforest canopy.	Rainforest	Escort / Platforming
Barramundi Run	Race barramundi upstream during spawning season in timed aquatic challenges.	Estuaries / Rivers	Racing
Cassowary Courier	Deliver items between ANPCs while carefully avoiding cassowary territory.	Rainforest	Delivery / Stealth
Croc Egg Watch	Protect nesting sites from predators in timed defense missions.	Estuaries / Mangroves	Tower Defense
Reef Clean-Up	Clear debris and crown-of-thorns starfish from coral sections to restore reef health.	Great Barrier Reef	Collection / Puzzle
Sugar Cane Maze	Navigate procedurally generated cane field labyrinths with randomized	Cane Fields	Exploration / Maze

Sidequest	Description	Biome	Type
	layouts each attempt.		
Night Market Dash	Speed-run through the Cairns Night Markets collecting ingredients for a recipe quest.	Urban Edge	Time Trial

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3. Characters

3.1 Wally — The Protagonist

Attribute	Detail
Species	Saltwater Crocodile (*Crocodylus porosus*)
Personality	Cheeky, brave, surprisingly gentle for a croc. Dry Australian humor. Loves a good swim. Unflappable in a crisis — unless there's a cassowary involved.
Visual Design	Vibrant green-olive scales, expressive yellow eyes, slightly oversized grin, compact muscular build optimized for platforming. Wears no clothing — natural croc design with stylized proportions for readability. Belly is a lighter cream-gold. Tail is slightly longer than anatomically accurate for gameplay purposes.
Abilities	Running, Jumping, Climbing, Swimming, Tail Whip (attack), CHOMP! (signature bite attack), Belly Slide, Death Roll (special move), Wall Jump, Belly Flop

3.2 ANPCs (Animated Non-Player Characters)

Name	Species	Role	Location	Personality
Baz the Barramundi	Barramundi	Quest Giver	Estuaries	Laconic and philosophical. Wise old fish who's seen it all. Speaks in slow, measured sentences. Loves a metaphor.
Coral the Sea Turtle	Green Sea Turtle	Reef Guide	Great Barrier Reef	Gentle and patient. Helps Wally navigate underwater sections. Deeply knowledgeable about reef ecology. Calming presence.
Bruce the Cassowary	Southern Cassowary	Rainforest Guardian	Daintree Rainforest	Grumpy but loyal. Initially hostile — must be won over through a quest. Fiercely territorial. Secretly has a soft side.
Shazza the Flying Fox	Spectacled Flying Fox	Aerial Scout	Rainforest / Urban	Energetic and chatty. Provides map intel and collectible locations. Never stops talking. Heart of gold beneath the noise.

Name	Species	Role	Location	Personality
Doug the Cane Toad	Cane Toad	Comic Relief Villain	Cane Fields / Estuaries	Bumbling but persistent. Leads the toad invasion with misplaced confidence. Thinks he's a mastermind. He is not.
Nev the Platypus	Platypus	Hidden Upgrade Provider	Freshwater Streams	Shy and cryptic. Found only in hidden freshwater areas. Provides secret upgrades. Speaks in riddles. May or may not exist — some ANPCs doubt his reality.
Maggie the Magpie	Australian Magpie	Town Informant	Urban Edge / Cairns City	Sarcastic and streetwise. Knows every shortcut, every back alley, every rubbish bin worth raiding. Sharp tongue, sharper beak.
Old Terry the Taipan	Coastal Taipan	Outback Sage	Atherton Tablelands	Ancient and mysterious. Gives cryptic hints about the final boss. Speaks with gravitas. Has witnessed centuries of change on the Tablelands. Unsettlingly calm.

3.3 Enemies & Bosses

Enemy	Type	Biome	Attack Pattern	Difficulty
Cane Toad Swarms	Basic Mob	Estuaries / Cane Fields	Hop toward player in groups of 3–6. Emit toxic puddles on defeat. Overwhelm through numbers.	Low
Bull Sharks	Patrol Enemy	Reef / Estuary	Patrol fixed routes in water. Charge when Wally enters detection radius. Fast, linear attack.	Medium
Crown-of-Thorns Starfish	Area Denial	Reef	Stationary. Damages on contact. Slowly expands coverage area. Must be removed from behind.	Low–Medium
Feral Pigs	Charging Enemy	Rainforest / Tablelands	Charges in straight line when alerted. Heavy damage and knockback. Vulnerable during recovery after charge.	Medium
Saltwater Croc Rival	Mini-Boss	Mangroves	Mirrors Wally's moveset. Uses CHOMP, Tail Whip, and submerge ambush. Must	High

Enemy	Type	Biome	Attack Pattern	Difficulty
			be baited into environmental traps.	
Giant Cane Toad King	Act 2 Boss	Cane Fields	Three-phase fight. Spawns toad swarms, creates toxic rain zones, ground-pound shockwaves. Weak point: belly during ground-pound recovery.	High
Box Jellyfish Colossus	Act 3 Boss	Great Barrier Reef	Massive jellyfish with sweeping tendril attacks. Creates current vortexes. Must strike bell during pulse phase. Full aquatic combat.	High
The Disruptor	Act 5 Final Boss	Atherton Tablelands	Mysterious entity at the source of the ecological disruption. Multi-phase fight across manifold geometry platforms. Adapts to player strategy. Tests mastery of all learned mechanics.	Very High

4. Gameplay Mechanics

4.1 Core Movement

Action	Input	Description	Animation State
Run	Left Stick / WASD	Standard horizontal movement with fluid acceleration curve. Wally transitions from walk to gallop based on stick pressure.	Walk → Run blend
Jump	A / Space	Variable height based on hold duration. Tap for short hop, hold for full leap. Double jump unlockable via upgrade.	Jump Launch → Apex → Land
Climb	Left Stick + Context	Cling to vines, tree trunks, and rock faces. Automatic grab when contacting climbable surfaces. Directional movement while climbing.	Climb Up / Climb Hang
Swim	Left Stick (in water)	Full 360-degree aquatic movement. Speed boost with tail thrust (RT/Shift). Wally is most agile in water — natural croc advantage.	Swim Forward / Swim Fast
Belly Slide	LT + Direction / Ctrl + Direction	Momentum-based slide on mud, wet surfaces, and slopes. Preserves and builds speed.	Belly Slide (loop)

Action	Input	Description	Animation State
		Can transition into other moves.	
Wall Jump	A / Space (at wall)	Bounce between vertical surfaces in tight spaces. Requires precise timing. Enables vertical traversal in urban and cave environments.	Wall Jump

4.2 Combat Mechanics

Move	Input	Range	Damage	Special Properties
CHOMP!	X / Left Click	Short (melee)	High	Signature melee attack. Forward lunge with jaw snap. Can **grab and throw** smaller enemies. Core offensive tool.
Tail Whip	Y / Right Click	360° sweep	Medium	Knocks back enemies and breaks destructible objects. Crowd control. Useful for clearing swarms.
Death Roll	LB+RB / Q (when Croc Meter full)	Grab range	Massive	Special move with cooldown. Wally grabs an enemy and performs a devastating axial spin. Requires full **Croc Meter** charge. Most powerful

Move	Input	Range	Damage	Special Properties
				move in Wally's arsenal.
Belly Flop	X / Left Click (airborne)	Area (below)	High	Aerial slam — ground-pound equivalent. Creates splash damage in water. Stuns nearby enemies on impact. Can trigger environmental switches.

4.3 Progression System

Croc Meter

The **Croc Meter** builds with successful attacks, combo chains, and collectible pickups. When full, it enables the devastating **Death Roll** special move. The meter is displayed as a subtle jaw icon in the HUD that fills with green energy. After use, the meter resets and must be recharged.

Scale Armor (Health System)

Wally's health is represented as **scale segments** along the screen edge. Each hit removes a segment. Collect **Scale Shards** found throughout levels to add permanent segments (base: 5 segments, maximum: 12). Health regenerates slowly when submerged in clean water.

Upgrades

Upgrade	Effect	Source
Tougher Scales	Reduces damage taken by 15% per tier (3 tiers)	Scale Shards collection milestone
Longer Lungs	Extends underwater	Reef exploration rewards

Upgrade	Effect	Source
	breath time by 30 seconds per tier (3 tiers)	
Sharper Teeth	Increases CHOMP! damage by 20% per tier (3 tiers)	Combat challenge completions
Faster Tail	Increases movement speed and swim speed by 10% per tier (3 tiers)	Time trial completions
Night Vision	Reveals hidden paths and collectibles in dark levels and underwater caves	Nev the Platypus (hidden character)

Collectibles

Collectible	Purpose	Frequency
Golden Mangoes	Currency for purchasing upgrades and cosmetics from ANPCs	Abundant (20–50 per level)
Scale Shards	Permanent health segment additions	Rare (1–2 per level, often hidden)
Croc Eggs	100% completion tracking items — one hidden in each level	One per level (well-hidden)
Postcards	Lore and regional facts about Cairns. Unlock gallery entries with real information about Far North Queensland	Moderate (3–5 per level)

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5. Level Design

5.1 Biome Overview

The Daintree Rainforest in Far North Queensland — visual reference for the rainforest biome

The Daintree Rainforest — visual reference for the dense canopy environment of Act 2.

Biome	Acts	Visual Theme	Key Hazards	Music Style	Unique Mechanic
Estuarine Wetlands	Act 1 (Tutorial)	Murky greens and browns; mudflats, tidal flows, mangrove fringe	Tidal surges, cane toad toxin puddles	Laid-back acoustic didgeridoo	Tidal water level changes — the environment shifts vertically on a timer, opening and closing paths
Daintree Rainforest	Act 2A	Dense canopy greens, filtered golden light, vertical layering	Cassowary charge zones, feral pig stampedes	Ambient jungle percussion	Vertical vine climbing and swinging — heavy vertical platforming through canopy layers
Sugar Cane Fields	Act 2B	Golden amber fields, harvest machinery, dusty atmosphere	Cane toad ambushes, harvester machines (instant kill)	Upbeat country folk	Procedural maze generation — cane rows create randomized labyrinths each playthrough
Mangrove Swamps	Act 2C	Tangled roots, brackish water, heavy atmosphere	Mud sinkholes, rival croc territory	Moody ambient bass	Root-network navigation and mud-slide mechanics — interconnected root tunnels as platforming routes

Biome	Acts	Visual Theme	Key Hazards	Music Style	Unique Mechanic
Great Barrier Reef	Act 3	Vivid coral colors, biolumines cent accents, crystal-clear water	Box jellyfish tendrils, current pulls, coral bleaching zones	Underwater electronica	Full aquatic movement and breath manageme nt — oxygen becomes a core resource
Cairns Urban Edge	Act 4	Tropical city, neon Night Markets, concrete and palm trees	Traffic hazards, storm drain currents	Tropical lo-fi beats	Stealth sections — Wally must avoid detection by humans while navigating urban spaces
Atherton Tablelands	Act 5	Volcanic lakes, cascading waterfalls, ancient formations	Ancient hazards, final boss arena	Epic orchestral	Crater lake boss arena with manifold geometry platforms — terrain shifts dynamicall y during the final fight

Great Barrier Reef underwater coral formations — visual reference for the Act 3 reef biome

Great Barrier Reef coral formations — inspiration for the vivid underwater world of Act 3.

5.2 Level Architecture — ButterflyFX Manifold System

Core Technology: ButterflyFX Manifold Architecture

All level geometry in CHOMP! is built on the ButterflyFX manifold surface system.

> This is the technological heart of the game and the foundation of the "Store Geometry Not Bytes" philosophy.

Levels are constructed on **ButterflyFX manifold surfaces** using foundational equations including **$z=xy$** and **$z=xy^2$**. These mathematical surfaces define the topological structure of every terrain mesh in the game.

- **Terrain meshes are procedurally generated** from manifold parameters, not stored as static geometry. A full level's terrain data can be encoded in kilobytes rather than megabytes.
- **"Store Geometry Not Bytes"** — level data is encoded as manifold parameters (seed values, curvature coefficients, boundary conditions), enabling dramatic file-size reductions across all platforms.
- **The manifold system enables smooth, organic terrain curves** that are natural to Australian landscapes — rolling mudflats, curved mangrove roots, undulating coral formations, and volcanic crater rims.
- **Procedural elements per biome:** Cane field mazes are generated from row-spacing parameters. Coral formations grow from branching manifold seeds. Vine networks are spanned across canopy anchor points. Root systems weave through manifold-defined soil layers.
- **Each biome has a unique manifold profile** that defines its geometric character — the Estuaries use gentle, low-frequency curves; the Reef uses high-frequency branching structures; the Tablelands use steep volcanic caldera profiles.

5.3 Level Flow Template

Each level in CHOMP! follows a consistent five-zone flow structure designed to teach, reward, challenge, and climax:

Zone	Purpose	Design Characteristics
1. Introduction Zone	Teach — safely introduce the level's new mechanic or hazard	Safe space. No enemies. Clear visual language. Player can experiment without penalty.
2. Exploration Zone	Reward — open area for collectibles, optional paths, ANPC encounters	Wide, branching layout. Multiple paths with varying difficulty. Hidden collectibles reward thorough exploration.
3. Challenge Zone	Test — escalating hazards and enemy encounters demand skill application	Linear progression through increasing difficulty. Combines new mechanic with existing skills. Checkpoint totems at key junctions.
4. Climax Zone	Excite — mini-boss fight or dramatic set piece	Arena or scripted sequence. Tests full mastery of the level's theme. Spectacular visual payoff. Rewarding victory moment.
5. Exit / Transition	Resolve — wind down, collect rewards, set up next level	Brief, safe. Summary of collectibles. ANPC dialogue teases next area. Auto-save at transition point.

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6. Hazards & Environmental Systems

CHOMP! features a rich ecosystem of environmental hazards drawn from real Far North Queensland conditions. Each hazard has a clear visual tell, a consistent behavior pattern, and at least one reliable counter — ensuring challenge without frustration.

Hazard	Biome(s)	Behavior	Damage	Counter
Tidal Surges	Estuaries	Periodic water level rise and fall on a predictable timer. Submerges lower platforms and opens upper routes.	Environmental (drowning if caught without air)	Ride the tide or find high ground. Watch tidal indicators on screen edge.
Cane Toad Toxin Puddles	Cane Fields / Estuaries	Toxic ground patches left by defeated toads or spawned by Doug. Persist for 10 seconds.	Damage over time (1 scale segment per 3 seconds of contact)	Jump over or belly slide past. Can be cleared with water splash from Belly Flop.
Box Jellyfish Tendrils	Reef	Drifting sting curtains that move with ocean currents. Semi-transparent — require attention to spot.	High (2 scale segments per hit)	Swim between gaps in the curtain or use current mechanics to push them aside.
Harvester Machines	Cane Fields	Moving industrial hazards that traverse cane rows on fixed paths. Loud audio and visual cues warn of approach.	Instant kill	Timing-based avoidance. Hide in gaps between cane rows. Audio and ground-shake cues provide warning.
Cassowary Charge Zones	Rainforest	Territorial attack lanes marked with visual indicators (scratched trees, feather markers). Bruce patrols	Heavy knockback + 2 scale segments	Stay out of marked zones. After befriending Bruce, these zones become safe passages.

Hazard	Biome(s)	Behavior	Damage	Counter
		until befriended.		
Mud Sinkholes	Mangroves	Quicksand-like traps that slow Wally, then deal damage. Visually distinct (darker, bubbling mud).	Slow debuff → 1 scale segment per 2 seconds	Belly slide to escape (momentum breaks free). Avoid by sticking to root platforms.
Stinger Season Currents	Reef	Seasonal jellyfish migration creates dense, impassable walls of stingers that move across the level.	Area denial (contact = high damage)	Wait for gap in migration wave or find alternate underwater route through coral tunnels.
Cyclone Events	Multiple biomes	Screen-wide wind and rain events that occur mid-level. Debris flies across screen. Strong lateral wind force.	Movement debuff + debris contact damage (1 segment)	Find shelter objects (overhangs, caves, thick canopy). Wait out the storm.
Feral Pig Stampedes	Rainforest / Tablelands	Charging herds triggered by entering their territory. Move in a wave from background to foreground.	Heavy damage (3 scale segments)	Climb to high ground (trees, rocks) to let the herd pass below.
Coral Bleaching Zones	Reef	Unstable, white-colored coral platforms that crumble underfoot. Visual cue: pale, cracking	Fall damage into deep water / hazard zone below	Move quickly across bleached sections. Don't linger. Look for healthy (colorful) coral

Hazard	Biome(s)	Behavior	Damage	Counter
		texture.		as safe platforms.
Storm Drain Currents	Urban	Water flow channels in Cairns storm drains that carry Wally downstream. Forced lateral movement.	Forced movement into hazards (context-dependent)	Grab ledges, pipes, and grates to resist the current. Swim against flow with Swim Fast.
UV Exposure	Open areas (multiple)	Time limit in direct sun — Wally overheats. Temperature gauge fills in exposed areas. Far North Queensland sun is no joke.	1 scale segment per 10 seconds of full exposure	Find shade (trees, overhangs, structures) or submerge in water to cool down.

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7. Art Direction

7.1 Visual Style

- **2.5D perspective with depth layers** — foreground action plane with parallax mid-ground and background layers creating rich environmental depth
- **Stylized realism** — recognizable Australian flora and fauna with slight cartoon proportions for gameplay readability. Animals are identifiable as their real species but with personality-driven exaggeration
- **Bold, saturated tropical color palette** — lush greens, ocean blues, coral pinks, golden cane, sunset oranges. Far North Queensland's palette is inherently vivid and the game leans into that

- **Dynamic lighting per biome:** Dappled filtered light in the rainforest canopy; underwater caustics on the reef; neon glow at the Night Markets; harsh, high-contrast outback sun on the Tablelands
- **Clean, minimal HUD:** Croc Meter displayed as a subtle jaw icon (fills with green energy). Scale Armor as segmented scales along the screen edge. Collectible counters fade in only when relevant. The world, not the UI, is the focus.

Cairns Night Markets — visual reference for the neon-lit urban biome in Act 4

Cairns Night Markets — reference for the tropical neon atmosphere of the Act 4 Urban Edge biome.

7.2 Color Palette by Biome

Biome	Primary	Secondary	Accent	Mood
Estuaries	Muddy Olive	Warm Brown	Golden Sun	Relaxed / Earthy
Rainforest	Deep Emerald	Moss Green	Filtered Gold	Mysterious / Lush
Cane Fields	Golden Amber	Burnt Sienna	Harvest Red	Warm / Nostalgic
Mangroves	Dark Teal	Root Brown	Murky Gold	Tense / Atmospheric
Reef	Vivid Coral	Ocean Blue	Bioluminescent Cyan	Vibrant / Magical
Urban	Concrete Gray	Neon Pink	Tropical Teal	Energetic / Modern
Tablelands	Volcanic Black	Waterfall Blue	Lava Orange	Epic / Dramatic

7.3 Animation Style Principles

- **Wally's animations prioritize weight and fluidity** — he's a crocodile, so movements should feel powerful but surprisingly agile. The gallop run has a low-belly lunge quality. The swim is a full-body sinusoidal wave.

- **Idle animations include natural behaviors:** slow blink, jaw gape (thermoregulation — real crocs do this), subtle tail sway, occasional head tilt toward nearby points of interest.
- **All ANPCs have unique idle loops** reflecting their species behavior — Baz drifts lazily, Shazza hangs upside-down and fidgets, Bruce scratches the ground, Maggie cocks her head and watches.
- **Environmental animations are procedural from manifold parameters:** Swaying sugar cane generated from wind-function overlays. Drifting jellyfish follow current-field vectors. Water surfaces are manifold-deformed planes. Rustling canopy responds to Wally's movement proximity.
- **Combat animations are punchy and satisfying** — CHOMP! has a brief anticipation frame followed by a rapid snap. Tail Whip has exaggerated follow-through. Death Roll uses motion blur on the spin. Impact frames use brief scale-flash for hit confirmation.

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8. 3D Rigging & Animation Specifications

8.1 Wally Rig Specifications

Component	Specification	Notes
Skeleton	42 bones minimum	Full bipedal-adapted crocodilian skeleton. Must support both quadrupedal locomotion and upright platforming poses.
Spine Chain	6 bones (head to tail tip)	Enables fluid serpentine motion for swimming and body undulation. IK/FK switchable for procedural vs. authored motion.

Component	Specification	Notes
Tail	8 bones with IK/FK blend	Critical for Tail Whip attacks, swimming propulsion, and idle sway. Must support rapid whip motion (16 frames) and gentle idle drift.
Jaw	3 bones (upper jaw, lower jaw, tongue)	Essential for CHOMP! animation. Upper jaw has slight upward motion on open. Tongue bone for grab/throw interactions.
Legs	4 bones per leg (shoulder/hip, upper, lower, foot) with IK	IK for ground contact. Legs must tuck cleanly for Belly Slide. Rear legs provide jump power; front legs for climbing grip.
Eyes	Individual bone per eye with look-at constraint	Eyes track points of interest (collectibles, enemies, ANPCs). Adds life and intelligence to Wally's expression.
Eyelids	2 bones per eye (upper lid, lower lid)	For blink cycles and expression. Slow blink = relaxed. Rapid blink = surprised. Narrowed = determined.
Toes/Claws	3 bones per foot	For grip animations on climbing surfaces. Claws spread on grab, retract on release. Visual feedback for climb state.
Blend Shapes	12 minimum facial expressions	Happy, angry, surprised, chomping, straining, sleeping, determined, hurt, smug, confused, excited, exhausted. Drive emotional readability.
Polygon Budget	8,000–12,000 tris	Mobile-friendly target. LOD system with 3 levels: Full (12K), Mid (6K), Far (3K). Silhouette must read at all LODs.
Texture Maps	Diffuse, Normal, Roughness, Emission	2048x2048 resolution. Emission map for subtle

Component	Specification	Notes
	(eyes only)	eye glow — gives Wally life in dark environments. PBR-compatible.

8.2 Animation Requirements

Animation	Frames	Loop	Priority	Notes
Idle Stand	60	Yes	Low	Includes blink cycle every 3–4 seconds with natural variation. Subtle breathing motion. Occasional jaw gape (thermoregulation behavior).
Idle Swim	48	Yes	Low	Gentle tail undulation. Limbs relaxed at sides. Slight vertical bob. Wally looks content — water is his element.
Run	24	Yes	High	Four-legged gallop with belly low to ground. Powerful rear-leg drive. Tail counterbalances. Feels heavy but fast.
Jump Launch	8	No	High	Powerful leg thrust from coiled position. Tail snaps upward.

Animation	Frames	Loop	Priority	Notes
				Explosive, satisfying takeoff.
Jump Apex	12	Yes	Medium	Brief hang with limbs spread. Tail extended for balance. This is where the player commits to landing direction.
Jump Land	6	No	High	Impact absorption through legs. Brief belly-low crouch. Dust/splash particles on impact. Quick recovery to idle or run.
Swim Forward	20	Yes	High	Full-body sinusoidal wave from snout to tail tip. Legs tucked. Elegant, powerful aquatic movement.
Swim Fast	16	Yes	High	Tail-powered burst. Higher frequency wave. Bubble trail. Speed lines. Used for tail thrust boost mechanic.
Climb Up	24	Yes	High	Alternating claw grip. Weight shifts visibly between

Animation	Frames	Loop	Priority	Notes
				limbs. Claws dig into surface with grip particles.
Climb Hang	30	Yes	Low	Clinging idle. Tail hangs and sways. Occasional claw readjustment. Wally looks around cautiously.
CHOMP Attack	12	No	Critical	Jaw snap with forward lunge. 2 frames anticipation, 3 frames snap, 7 frames recovery. Punchy and immediate. Screen shake on hit.
Tail Whip	16	No	Critical	Full 360° sweep. Wally plants front legs and rotates rear body. Exaggerated tail follow-through. Knockback feedback on hit.
Death Roll	36	No	Critical	Grab (4 frames) + rapid axial rotation (24 frames) + release (8 frames). Motion blur on spin. Water splash if aquatic. Most

Animation	Frames	Loop	Priority	Notes
				dramatic animation in the game.
Belly Slide	20	Yes	High	Legs tucked to sides. Belly flat on surface. Momentum-based — body flexes with terrain contour. Mud/water spray particles.
Belly Flop	14	No	High	Aerial descent with belly presented. Impact frame with large splash/shockwave. Brief bounce recovery. Environmental reaction (waves, debris).
Hurt	8	No	Critical	Flinch with scale flash (white flash on damaged scales). Brief invulnerability window. Slight recoil in hit direction. Clear damage feedback.
Defeat	48	No	Critical	Dramatic collapse — legs give way, body tips, belly up. Classic crocodile "playing dead"

Animation	Frames	Loop	Priority	Notes
				posture. Comedic rather than distressing. Eyes spiral.
Victory	36	No	Low	Happy jaw gape + enthusiastic tail wag. Wally rises slightly on hind legs. Celebratory energy. Used at end of boss defeats and level completion.
Wall Jump	10	No	High	Kick off surface with rear legs. Brief contact frame. Directional launch. Tight, responsive timing required.

8.3 ANPC Rig Guidelines

Specification	Requirement
Skeleton Type	Species-appropriate skeleton per ANPC. Fish (Baz): spine-chain only (12 bones). Birds (Bruce, Maggie, Shazza): wing bones + leg IK (24–30 bones). Mammals/Reptiles (Nev, Old Terry, Doug): standard quadruped/biped (18–28 bones).
Idle Variations	Minimum 2 idle variations per ANPC. Each reflects species-specific natural behavior. Randomized selection with weighted probability.
Dialogue State	Subtle head movement + mouth sync.

Specification	Requirement
	4 viseme shapes minimum (open, closed, wide, narrow). Head tilts and eye darts during conversation.
Blink Cycle	All ANPCs must have a natural blink cycle with randomized timing (2–5 second intervals). Nictitating membrane for reptile/amphibian ANPCs.
Polygon Budgets	Major ANPCs (Baz, Coral, Bruce, Shazza, Doug): 4,000–6,000 tris. Minor ANPCs (Nev, Maggie, Old Terry): 1,500–3,000 tris. LOD support for all.

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9. Audio Design

9.1 Music Direction

CHOMP! features a biome-driven adaptive music system that blends authentic Australian instrumentation with modern game scoring techniques. Music is not background — it is an active participant in the gameplay experience.

- **Each biome has a unique musical theme** that establishes place and mood immediately. The Estuaries open with laid-back didgeridoo drones. The Reef pulses with underwater electronica. The Tablelands crescendo with epic orchestral scoring.
- **Adaptive music system:** Each biome track is composed in intensity layers. During exploration, only ambient layers play. Entering combat adds percussion and melodic layers. Boss encounters activate full orchestration. Transitions are seamless and musical.
- **Core instruments:** Didgeridoo, clap sticks, acoustic guitar, tropical percussion (bongos, steel drums, rain sticks), electronic synthesizers for underwater and urban biomes, full orchestral ensemble for boss encounters and the Tablelands finale.

- **Boss music:** Full orchestral + electronic hybrid scoring with rising intensity across each boss phase. Phase transitions are marked by musical shifts — tempo increases, key changes, and new instrument entries.

9.2 Sound Effects

Category	Examples	Style
Wally Movement	Footsteps on mud, rock, water, and wood surfaces; splash entry/exit; belly slide friction; climb grip scrape; swimming strokes	Realistic with slight stylization. Surface-dependent footstep system. Water interactions are particularly detailed — Wally is an aquatic creature and it should sound like it.
Combat	CHOMP! jaw snap; tail swoosh and impact thud; death roll splash and crunch; belly flop impact and wave; enemy hit reactions; scale armor clink on block	Punchy and satisfying. CHOMP! is the signature sound — a deep, resonant snap that feels powerful. Every hit should feel impactful. Brief bass rumble on Death Roll activation.
Environmental	Rain (light drizzle to cyclone downpour); wind through cane fields; water flow (creek, river, tidal); bird calls; insect chorus (cicadas, crickets); thunder; rustling canopy; underwater bubbles	Authentic Cairns ambient audio. Sourced or inspired by real Far North Queensland soundscapes. Biome transitions blend environmental audio gradually. Day/night variations.
UI	Menu select; collectible pickup (Golden Mango chime, Scale Shard crystal tone); upgrade unlock fanfare; quest complete stinger; checkpoint totem activation; map open/close	Clean, tropical-themed chimes. Consistent audio family. Golden Mango pickup is a bright, ascending marimba note. Scale Shard is a deeper, resonant crystal tone.
ANPC Voices	Baz: deep, slow bubbles; Coral: gentle chirps; Bruce: aggressive clucks	Stylized animal sounds with personality — not voice-acted dialogue, but

Category	Examples	Style
	and booms; Shazza: rapid squeaks; Doug: comical croaks; Nev: quiet chatters; Maggie: sharp warbles; Old Terry: low hiss	expressive vocalizations that convey emotion and character. Each ANPC is identifiable by sound alone.

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10. Technical Specifications

10.1 Engine & Framework

ButterflyFX — Manifold Geometry Engine

CHOMP! is built entirely on the **ButterflyFX** manifold geometry engine — a proprietary technology developed by KensGames. ButterflyFX uses manifold mathematical surfaces as the foundational building blocks for all level geometry, enabling the "Store Geometry Not Bytes" approach that defines the game's technical identity.

- **Manifold surfaces** (**$z=xy$**, **$z=xy^2$**) serve as foundational level geometry equations. All terrain is derived from these mathematical relationships.
- **Procedural generation** from manifold parameters handles terrain, flora, environmental features, and dynamic level elements like cane field mazes and coral formations.
- **Cross-platform deployment:** Steam (PC/Mac), Google Play (Android), with architecture designed for console expansion.
- **Rendering pipeline:** Forward rendering with per-biome post-processing profiles. Underwater caustics, volumetric fog (rainforest), neon bloom (urban), and heat shimmer (outback) are biome-specific effects.

10.2 Performance Targets

Platform	Resolution	Frame Rate	Draw Distance	Max Entities
PC (Steam)	1920x1080 / 4K	60 FPS	100m	200
Mobile (Google Play)	1280x720 / 1920x1080	30 FPS	50m	100
Future Console	1920x1080 / 4K	60 FPS	100m	200

10.3 Save System

- **Auto-save** triggers at level transitions and checkpoint totems — carved crocodile totems placed at key junctions throughout each level. Visual and audio feedback confirms save.
- **3 save slots** — each independently tracked with playtime, completion percentage, and last-saved location preview.
- **Cloud save sync** across platforms — a player can progress on PC and continue on mobile seamlessly.
- **Save data includes:** Level progress, collectibles found (Golden Mangoes, Scale Shards, Croc Eggs, Postcards), upgrades unlocked, quest states, ANPC relationship flags, play time, and difficulty setting.

10.4 Accessibility Features

Feature	Description
Colorblind Modes	Three modes: Protanopia, Deuteranopia, Tritanopia. All gameplay-critical color cues (hazard indicators, collectible types, health states) have alternate palettes and shape-based secondary indicators.

Feature	Description
Adjustable Text Size	Three text size options (Standard, Large, Extra Large) for all UI text, dialogue, and collectible descriptions.
Remappable Controls	Full control remapping on all platforms. Keyboard, gamepad, and touch layouts are independently configurable.
Visual Audio Indicators	All audio-critical gameplay cues (hazard warnings, enemy alerts, environmental events) have corresponding visual indicators — screen-edge flashes, directional arrows, or icon prompts.
Difficulty Settings	**Tourist** (Easy): More Scale Armor segments, slower hazards, generous checkpoints, enemy damage reduced. **Local** (Normal): Standard experience as designed. **Croc Hunter** (Hard): Fewer health segments, faster hazards, limited checkpoints, aggressive enemy AI.

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11. Monetization & Release Strategy

Business Model

CHOMP! is a **premium game — one-time purchase** on all platforms. There are no microtransactions, no loot boxes, no battle passes, and no pay-to-win mechanics. The player buys the game and gets the full experience. Period.

Post-Launch Content

DLC	Description	Scope
Cape Tribulation Expansion	New biome extending the Daintree into Cape Tribulation — where the rainforest meets the reef.	5–8 new levels, 2 new ANPCs, 1 boss

DLC	Description	Scope
	New levels, ANPCs, and a unique amphibious boss encounter.	
Torres Strait Islands Expansion	Open-water archipelago biome with island-hopping mechanics, tidal navigation, and Torres Strait Islander cultural elements (developed in consultation with community advisors).	6–10 new levels, 3 new ANPCs, 1 boss

Release Strategy

- **Platform launches:** Steam, Google Play, and major game platforms
- **Review outreach:** Targeted submissions to game critics, indie-focused outlets, and Australian gaming media
- **Community hub:** KensGames.com serves as the central hub, portfolio showcase, devlog platform, and community gathering point
- **Marketing pillars:** Authentic Australian setting (unique in the platformer space), ButterflyFX technology showcase, charming character cast, humor-driven tone

Atherton Tablelands waterfall — visual reference for the Act 5 finale environment

Atherton Tablelands waterfall — reference for the epic natural formations of the Act 5 finale.

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12. Appendices

Appendix A: Cairns Regional Reference Guide

The following table maps real-world Cairns and Far North Queensland elements to their in-game representations. CHOMP! is committed to regional authenticity — every hazard, creature, and landmark is drawn from the real place.

Element	Real-World Reference	In-Game Usage
Saltwater Crocodiles	Largest living reptile. Common in FNQ estuaries, rivers, and coastal areas. Apex predator.	Wally (protagonist). Rival croc mini-boss. Checkpoint totems carved in croc form. Core identity of the game.
Great Barrier Reef	World's largest coral reef system. UNESCO World Heritage Site off the Queensland coast.	Entire Act 3 biome. Coral platforms, reef predators, underwater exploration. Environmental theme of coral health.
Daintree Rainforest	One of Earth's oldest rainforests (~180 million years). Dense tropical canopy north of Cairns.	Act 2A biome. Canopy platforming, vine mechanics, vertical level design. Bruce the Cassowary's territory.
Sugar Cane Industry	Major FNQ agricultural industry. Vast cane fields dominate the coastal lowlands between Cairns and Innisfail.	Act 2B biome. Procedural cane mazes, harvester machine hazards. Cultural backdrop for rural levels.
Cane Toads	Introduced pest (1935). Highly invasive. Toxic. A notorious ecological disaster in Australian history.	Doug the Cane Toad (villain). Cane Toad Swarms (basic enemy). Toxin puddle hazards. Toad Roundup sidequest. Giant Cane Toad King boss.
Cassowaries	Southern Cassowary — endangered, territorial, large flightless bird. Can be aggressive. Found in Daintree and surrounding rainforests.	Bruce the Cassowary (ANPC). Cassowary Charge Zones (hazard). Cassowary Courier sidequest.
Box Jellyfish	Highly venomous marine stinger. "Stinger season" (October–May) is a real annual hazard in FNQ waters.	Box Jellyfish Tendrils (hazard). Stinger Season Currents (area denial). Box Jellyfish Colossus (Act 3 boss).
Flying Foxes	Spectacled Flying Fox — large fruit bat. Colonies roost in trees around Cairns. Important pollinator. Sometimes controversial in urban	Shazza the Flying Fox (ANPC aerial scout). Flying Fox Rescue sidequest.

Element	Real-World Reference	In-Game Usage
	areas.	
Cairns Night Markets	Popular tourist attraction on Abbott Street. Open nightly. Food, crafts, souvenirs. Iconic Cairns experience.	Act 4 level setting. Night Market Dash sidequest (speed-run time trial). Neon-lit urban platforming environment.
Atherton Tablelands	Elevated plateau west of Cairns. Volcanic crater lakes (Lake Eacham, Lake Barrine), waterfalls, cooler climate.	Act 5 biome. Final boss arena set in volcanic crater. Waterfall traversal. Old Terry the Taipan's territory.
Barramundi Fishing	Iconic Australian sport fish. Prized catch in FNQ rivers and estuaries. Cultural significance in the region.	Baz the Barramundi (ANPC quest giver). Barramundi Run sidequest (racing/time trial).
Cyclone Season	November–April tropical cyclone season. Severe weather events are part of life in FNQ.	Cyclone Events (multi-biome hazard). Screen-wide weather effects, wind mechanics, shelter gameplay.
Taipan Snakes	Coastal Taipan — highly venomous snake found in FNQ. Reclusive but dangerous.	Old Terry the Taipan (ANPC outback sage). Provides cryptic hints about the final boss. Ancient, wise character.
Feral Pigs	Invasive species. Destructive to native habitats. Found in rainforest and rural areas across FNQ.	Feral Pigs (charging enemy). Feral Pig Stampedes (hazard). Environmental commentary through gameplay.
Magpies	Australian Magpie — intelligent, territorial bird. Famous for swooping during breeding season. Found throughout urban and suburban Australia.	Maggie the Magpie (ANPC town informant). Sarcastic urban guide for the Cairns city levels.

Appendix B: Control Mapping

Keyboard & Mouse (PC — Default)

Action	Key/Button
Move	WASD
Jump	Space
CHOMP! Attack	Left Mouse Button
Tail Whip	Right Mouse Button
Belly Slide	Ctrl + Direction
Death Roll	Q (when Croc Meter full)
Swim Fast / Tail Thrust	Shift (in water)
Interact / Talk	E
Map	M
Pause / Menu	Esc

Gamepad (Default)

Action	Button
Move	Left Stick
Camera (where applicable)	Right Stick
Jump	A / Cross
CHOMP! Attack	X / Square
Tail Whip	Y / Triangle
Belly Slide	LT / L2 + Direction
Death Roll	LB + RB / L1 + R1
Swim Fast / Tail Thrust	RT / R2 (in water)
Interact / Talk	B / Circle
Map	Select / Touchpad
Pause / Menu	Start / Options

Touch Controls (Mobile — Default)

Action	Input
Move	Virtual joystick (left side)
Jump	A button (right side)
CHOMP! Attack	X button (right side)
Tail Whip	Y button (right side)
Belly Slide	Swipe down on joystick
Death Roll	Double-tap Croc Meter icon (when full)
Swim Fast	Double-tap joystick direction (in water)
Interact / Talk	Context button (appears near ANPCs/objects)
Map	Map icon (top right)
Pause / Menu	Pause icon (top right)

Appendix C: Glossary

Term	Definition
ANPC	**Animated Non-Player Character.** KensGames terminology for NPCs with full species-appropriate animation rigs, behavioral idle loops, and expressive dialogue systems. Distinguished from static NPCs by their living, breathing presence in the world.
ButterflyFX	Proprietary manifold geometry engine developed by KensGames. Uses mathematical surface equations to define and generate level geometry procedurally. Core technology powering CHOMP!
Manifold Architecture	Level design methodology where terrain and environmental geometry are defined by manifold surface parameters (e.g., $z=xy$, $z=xy^2$) rather than stored as static mesh data. Enables the "Store Geometry Not

Term	Definition
	Bytes" approach.
Store Geometry Not Bytes	Core design philosophy of CHOMP! and ButterflyFX. Level geometry is encoded as compact mathematical parameters rather than large static mesh files, dramatically reducing file sizes and enabling real-time procedural generation.
Croc Meter	Combat resource gauge that fills through successful attacks, combos, and collectible pickups. When full, enables Wally's devastating Death Roll special move. Displayed as a jaw icon in the HUD.
Scale Armor	Wally's health system. Represented as segmented crocodile scales along the screen edge. Each hit removes a segment. Expanded permanently by collecting Scale Shards. Base: 5 segments, maximum: 12.
CHOMP!	Wally's signature melee attack — a powerful jaw snap with forward lunge. Short range, high damage. Can grab and throw smaller enemies. The move that gives the game its name.
Death Roll	Wally's ultimate special move. A devastating grab-and-spin attack inspired by real saltwater crocodile hunting behavior. Requires full Croc Meter charge. Most damaging move in Wally's arsenal.
Manifold Profile	A biome-specific set of manifold parameters that defines the geometric character of a level's terrain. Each biome in CHOMP! has a unique profile — e.g., gentle curves for estuaries, steep caldera shapes for the Tablelands.
Checkpoint Totem	In-world save point represented as a carved crocodile totem. Activating a totem triggers auto-save and serves as a respawn point. Placed at key junctions throughout each level.
Golden Mango	Primary currency collectible. Abundant

Term	Definition
	throughout levels (20–50 per level). Used to purchase upgrades and cosmetics from ANPCs.
Postcard	Lore collectible containing real information about Cairns, Far North Queensland, and Australian wildlife and culture. Unlocks gallery entries. 3–5 hidden per level.

CHOMP! — Wally the Cairns Crocodile

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KensGames (kensgames.com) — Powered by ButterflyFX

Creative Director: Kenneth Bingham

"Store Geometry Not Bytes"

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